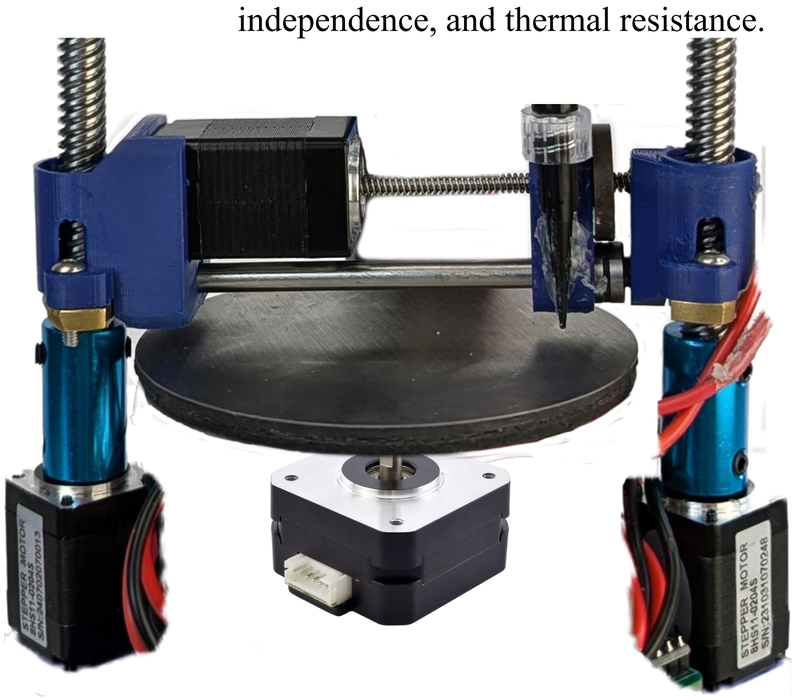
CubeSat DR3AM: A 4U Demonstrative Robust 3D Additive Manufacturing System

Spencer Stowell, stowel22@byu.edu; Casey Gooch, akcaseygooch@gmail.com

Overview



DR3AM was engineered to demonstrate volumetric efficiency and survivability of a **4U CubeSat 3D printer** under rocket flight conditions. Developed for the 2024 Space Dynamics Laboratory Payload Challenge, it competed against over 150 collegiate teams worldwide and earned **first place** for its innovative demonstration of additive manufacturing in flight-relevant conditions. Secondary design objectives included energy efficiency, gravity independence, and thermal resistance.



Motion Architecture

After considering **cartesian** and **delta** configurations, a **cylindrical configuration** was selected for *space and part-efficiency,* as well as *mechanical robustness*, while providing *three degrees of freedom*.

DOF's:

R: Nema 8 stepper motor + gantry, mounted crossways

Z: Dual Nema 8 stepper motors, synchronized from the same motor controller

θ: Nema 17 pancake stepper, mounted directly to the plate for continuous rotation

The **kinematics** of the build plate and physical constraints of stepper motors can lead to a **singularity** near the center and low resolution at the edge, reducing the capabilities of this system.

DR3AM



Preflight Timelapse

Postflight Timelapse

Extrusion Architecture

DR3AM implements a **Direct-Ink-Write** system, where photocuring resin is extruded through a nozzle and cured by UV exposure.

- Significantly **reduces power** & waste heat
- 30W vs 70W (average FDM) total power
- 3W vs 30W to deposit material



Resin is contained in a **20 mL syringe** with an embedded lead screw. Actuated with a stepper motor and 19:1 planetary gearbox, this enables DR3AM to extrude resin at a **similar resolution** to a standard 3D printer (30-40 steps/mm^3). 395 nm high-power LED's cure the resin as it extrudes.

Electronics

Computer:
- Raspberry Pi 4B w/ Klipper

Sensors:
- Accelerometer (launch detection)
- Switchless homing

Power electronics:
- Eight UV LEDs (4S2P)
- Two 6.4V 6 Ah LiFePO4 batteries (2S)

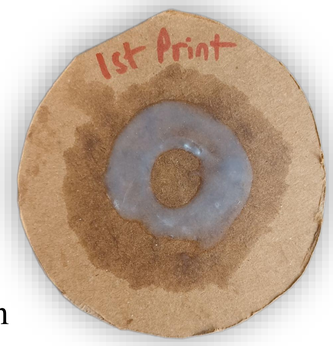
Microcontrollers:
- RPi Pico (stepper control)
- Arduino Nano (safety measures)

Actuators:
- Four Nema 8 steppers
- Nema 17 pancake stepper
- 12V solenoid safety valve (resin, fail close)

Results & Conclusions

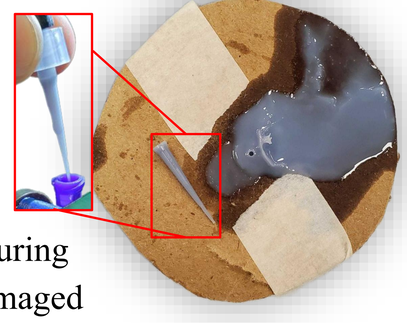
- Successful flight to **10K ft** on an L3 rocket
- Returned with broken zip ties, **bent radial gantry**, and misaligned z-axes
- **Permanently affected** the accuracy of the gantry

Timing of damages - during motor burn, chute deployment, or landing - is not currently known



Test Print Results

The DIW system needs **significant refinement** to be feasible:
- Reduce undesirable UV overexposure
- Improve extrusion consistency
- Increase radial gantry strength



2025 team experienced dynamic effects during launch and V2 prototype was severely damaged

Resin Capacity	Print Volume	Volumetric Efficiency	Operating Voltage	Power Consumption	UV Wavelength
20 mL	**700 cm^3**	**17.6%**	**12.8V**	**30W**	**395 nm**

Sponsors: BYU Brigham Young University; Weidman Center; BYU Engineering; onshape; US Synthetic a ChampionX company; ProtoSpace MFG; Utah NASA Space Grant Consortium

Team Members: Spencer Stowell, Casey Gooch, Taylor Anderton
Faculty Advisor: Dr. David Fullwood